TradifyERP

Knowledge Transfer Document

Developer & Administrator Guide

System Version:	TradifyERP v1.0
Total Modules:	12 Modules
Database:	PostgreSQL / SQLite
Framework:	Django 5.2.8
Python Version:	3.14
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Tradify Global
Enterprise Resource Planning System

1. System Overview

This document provides comprehensive knowledge transfer for TradifyERP developers and administrators. It covers system architecture, coding conventions, deployment procedures, and operational best practices.

2. Technology Stack

Component	Technology	Version
Backend Framework	Django	5.2.8
Programming Language	Python	3.14
Database	PostgreSQL / SQLite	15+ / 3.x
Frontend Framework	Bootstrap	5.3
Template Engine	Django Templates	5.2.8
Authentication	Django Allauth	Latest
Forms	Crispy Forms	Latest
Task Queue	Celery (Optional)	5.x

3. Project Structure

```
TradifyERP/
■■■ TradifyERP/
■ ■■■ __init__.py
■ ■■■ settings.py # Main configuration
■ ■■■ urls.py # Root URL configuration
■ ■■■ wsgi.py # WSGI application
■ ■■■ asgi.py # ASGI application
■■■ accounting/ # Core accounting module
■ ■■■ models.py # 29 models
■ ■■■ views.py # Business logic
■ ■■■ urls.py # URL routing
■ ■■■ forms.py # Form definitions
■ ■■■ templates/ # HTML templates
■■■ hr/ # HR module
■■■ manufacturing/ # Manufacturing module
■■■ sales/ # Sales module
■■■ inventory/ # Inventory module
■■■ procurement/ # Procurement module
■■■ ap/ # Accounts Payable
■■■ ar/ # Accounts Receivable
■■■ crm/ # CRM module
■■■ projects/ # Project management
■■■ cash_management/ # Cash management
■■■ maintenance/ # Site maintenance
■■■ templates/ # Global templates
■■■ static/ # Static files
■■■ manage.py # Django management
■■■ requirements.txt # Python dependencies
```

4. Database Schema Overview

The database consists of 122+ tables organized by module. All modules share the Entity model for multi-company support. Key relationships include:

• Entity → All modules (one-to-many)

• Account → LedgerEntry (one-to-many)

• VendorInvoice → VendorPayment (one-to-many)

• Customer → SalesOrder → SalesInvoice (cascading)

• Product → BOM → WorkOrder (manufacturing chain)

• Employee → Payroll → PayrollItem (HR hierarchy)

• Project → Task → TimeEntry (project tracking)

5. Key Business Processes

Order to Cash (O2C):

1. CRM Lead → Opportunity

2. Sales Quote → Sales Order

3. Sales Invoice (Revenue Recognition)

4. AR Collection → Bank Deposit

5. GL Posting (automatic)

Procure to Pay (P2P):

1. Purchase Requisition

2. RFQ → Purchase Order

3. Goods Receipt → Stock

4. Vendor Invoice (3-way match)

5. AP Payment → Bank

6. GL Posting (automatic)

Manufacturing Process:

1. Sales Order → Production Planning

2. BOM → Work Order

3. Material Requisition → Inventory

4. Production Execution

5. Quality Control
6. Finished Goods → Stock
7. Cost Accounting → GL

6. Security & Access Control

TradifyERP implements role-based access control (RBAC) with the following features:

• Role-based permissions (View, Create, Edit, Delete, Approve)

• Digital signatures for critical transactions

• Approval workflows with multi-level authorization

• Audit trail for all financial transactions

• Period locks to prevent backdating

• User action logging (who, what, when)

• Session management and timeout controls